TESLA COACHING CLASSES

Science and Technology – Part 1

Time: 2 Hours	Standard: Class 10	Total Marks: 40

Part Syllabus Test - 05 (PST-05)

Student Name: _____ **Date:** _____

CHAPTERS COVERED & WEIGHTAGE

CHAPTER 1	Gravitation	20.7%
CHAPTER 2	Periodic Classification of Elements	26.1%
CHAPTER 6	Refraction of Light	28.7%
CHAPTER 7	Lenses	24.5%

Q.1(A) **Choose the correct alternative and rewrite the sentences:** **[5 Marks]**

i) A _____ is necessary to change the speed as well as the direction of motion of an object.

(a) force **(b)** inertia

(c) momentum **(d)** motion

ii) X and Y are two elements having similar properties which obey Newlands's Law of Octaves. The minimum and maximum number of elements in between X and Y respectively are _____.

(a) 6 and 8 **(b)** 7 and 15

(c) 8 and 14 **(d)** 6 and 13

iii) If the refractive index of air with respect to glass is $2/3$. What is the refractive index of glass with respect to air?

(a) $2/3$ **(b)** $3/2$

(c) $1/3$ **(d)** $1/2$

iv) Arrange the following elements in order of their decreasing metallic character. Na, Si, Cl, Mg, Al

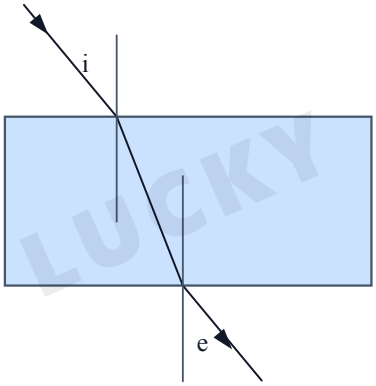
(a) Cl > Si > Al > Mg > Na **(b)** Na > Mg > Al > Si > Cl

(c) Na > Al > Mg > Cl > Si **(d)** Al > Na > Si > Ca > Mg

v) In the following diagram if $\angle i = 40^\circ$, then $\angle e =$ _____?

(a) 50° **(b)** 40°

(c) 60° **(d)** 90°



LUCKY

Q.1(B)	Answer the following sub-questions:	[5 Marks]

i) **Find the correlation:**
Two elements in the same position : demerit of Newland's octaves : : place for Isotopes : _____

ii) **Find the correlation:**
${}_2n_1$: Refractive index of medium 2 with respect to medium 1 : : ${}_1n_2$: _____

iii) **Right or wrong sentence:**
The CGS unit of G is $\mathrm{dyne} \cdot \mathrm{cm}^2/\mathrm{g}^2$.

iv) **State whether Right or Wrong:**
The speed of light varies in different media.

v) **Write the name.**
The distance between focus and optical centre.

Q.2(A)	Give scientific reasons (Any TWO):	[4 Marks]

i) The value of gravitational acceleration (g) is taken to be $-g$ when studying the motion of an object thrown upwards in a straight line.

ii) The bottom of a pond appears raised.

iii) The movie cannot be enjoyed if seat of a viewer is too close to the screen in the cinema.

Q.2(B)	Answer the following questions (Any THREE):	[6 Marks]

i) **Distinguish between:**
Universal gravitational constant – Gravitational acceleration of earth.

Universal gravitational constant	Gravitational acceleration of earth

ii) What is refraction of light?

iii) What physical and chemical properties of elements did Mendeleev consider in compiling the periodic table?

iv) Solve the following example.
Speed in first medium, $v_1 = 3 \times 10^8\ \mathrm{m/s}$
Speed in second medium, $v_2 = 2 \times 10^8\ \mathrm{m/s}$
Then ${}_2n_1 = ?$

v) **<u>Distinguish between:</u>**
Myopia - Presbyopia

Myopia	Presbyopia

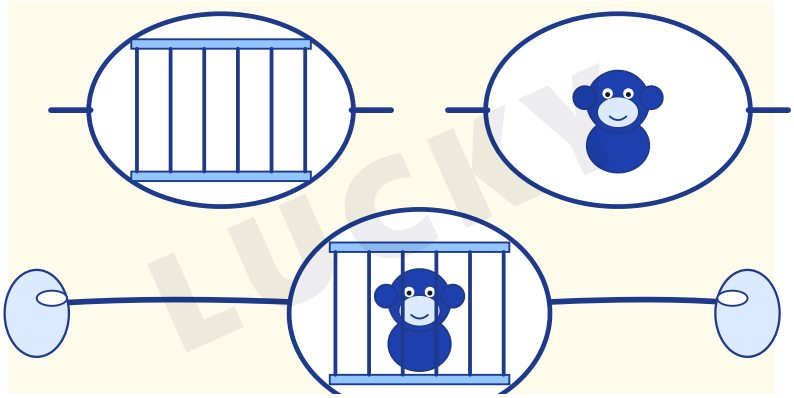
Q.3	Answer the following questions (Any FIVE):	[15 Marks]

i) The mass of planet 'X' is four times that of the earth, and its radius is double the radius of the earth. The escape velocity of a body from the earth is $11.2 \times 10^3\ \mathrm{m/s}$. Find the escape velocity of a body from the planet 'X'.

ii) What challenges did Mendeleev face in following periodic law?

iii) If the absolute refractive index of glass and water are $3/2$ and $4/3$ respectively, what is the refractive index of glass with respect to water?

iv) **<u>Identify and explain concepts given in this diagram.</u>**



v) The mass of the earth is $6 \times 10^{24}\ \mathrm{kg}$. The distance between the earth and the sun is $1.5 \times 10^{11}\ \mathrm{m}$. If the gravitational force between the two is $3.5 \times 10^{22}\ \mathrm{N}$, what is the mass of the sun? Use $G = 6.7 \times 10^{-11}\ \mathrm{N\ m^2\ kg^{-2}}$

vi) Explain structure of modern periodic table in short.

vii) If the absolute refractive index of water is 1.36, find the speed of light in water. (Speed of light in vacuum = $3 \times 10^8\ \mathrm{m/s}$.)

viii) $5\ \mathrm{cm}$ high object is placed at a distance of $10\ \mathrm{cm}$ from a converging lens of focal length of $20\ \mathrm{cm}$. Determine the position, size and type of the image.

Q.4	Answer the following question (Any ONE):	[5 Marks]

i) XY compound is formed by X (Atomic number 11) and Y (Atomic number 17), then answer the questions given below.

a. Determine the position of the element X and Y periodic table.

b. Which type of elements X and Y are metals, nonmetals or metalloids?

c. From which block the elements X and Y are?

d. Determine the electronic configuration and valency of these elements.

ii) <u>Choose the correct option from the bracket and complete the stanza.</u>

(colour blind, actual, conical, light-sensitive, rodlike, colours)

The retina in our eyes is made up of many _____ cells. These cells are shaped like a rod and like a cone. The _____ cells respond to the intensity of light and give information about the brightness or dimness of the object to the brain. The _____ cells respond to the colour and give information about the colour of the object to the brain. Brain processes all the information received and we see the _____ image of the object. Rod like cells respond to the faint light also but _____ cells do not. Some people lack conical cells responding to certain colours. These persons cannot recognize those colours or cannot distinguish between different _____. These persons are said to be _____.

ALL THE BEST

DESIGNED BY YOUR MENTOR – LUCKY BHAIYA